Software Engineering
Assignment-5
TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

Which of the following statements about modularity in software design are correct?

a. Modularity involves decomposing a problem into a set of modules.
b. Modules in a modular design are intended to be largely independent of one another.
c. Modularity is based on the divide and conquer principle.
d. Modularity always increases interdependence among modules.
e. Modularity is considered an important attribute of good software design.
f. Modularity eliminates the need for system integration.

Correct Answer: a. Modularity involves decomposing a problem into a set of modules.
b. Modules in a modular design are intended to be largely independent of one another.
c. Modularity is based on the divide and conquer principle
e. Modularity is considered an important attribute of good software design

Detailed Solution:

Modularity decomposes a problem into modules, promotes independence (low coupling), and applies the divide-and-conquer principle—all hallmarks of good design. It does not increase interdependence (d)—it reduces it—and it does not eliminate system integration (f); integration is still needed, though it becomes easier.

QUESTION 2:

Which of the following statements regarding coupling are correct?

a. Coupling refers to the degree of dependence among software components.
b. Highly coupled systems have many interdependencies among components.
c. High coupling is generally preferred in good software design.
d. In highly coupled systems, modifying one component may affect several other components.
e. Coupling and modularity are completely unrelated concepts.
f. Tightly coupled modules are easier to maintain and modify.

Correct Answer: a. Coupling refers to the degree of dependence among software components.
b. Highly coupled systems have many interdependencies among components
d. In highly coupled systems, modifying one component may affect several other components

Detailed Solution:

Coupling measures dependence between components; high coupling means many interdependencies, so changes ripple widely—making maintenance harder. High coupling is not preferred (c), coupling and modularity are closely related (e), and tightly coupled modules are harder to maintain (f), not easier.

QUESTION 3:

Which of the following statements correctly describe cohesion in software design?

a. Cohesion refers to how strongly the elements within a module are related to one another
b. High cohesion generally improves maintainability and readability of software
c. Cohesion measures the level of dependency between different modules
d. Functional cohesion is considered one of the strongest forms of cohesion
e. Low cohesion is usually preferred because it increases module flexibility
f. Cohesion is concerned with the internal organization of a module

Correct Answer: a. Cohesion refers to how strongly the elements within a module are related to one another
b. High cohesion generally improves maintainability and readability of software
d. Functional cohesion is considered one of the strongest forms of cohesion
e. Cohesion is concerned with the internal organization of a module

Detailed Solution:

Cohesion measures how strongly elements within a module are related—high cohesion improves maintainability and readability, and functional cohesion is among the strongest types. Cohesion is not about dependency between modules (that's coupling), and low cohesion is not preferred—it makes modules harder to understand and maintain.

QUESTION 4:

What does “common coupling” mean?

a. Two modules share global data
b. Modules communicate only through parameters
c. Modules rely on shared logic
d. Modules share a common interface
e. Modules share code
f. Modules share no data, as is the common practice

Correct Answer: a. Two modules share global data

Detailed Solution:

Common coupling occurs when two or more modules access the same global data area (e.g., global variables, shared memory), creating hidden dependencies and making maintenance difficult. It does not refer to parameter passing (b), shared logic (c), common interfaces (d), code sharing (e), or sharing no data (f)—those describe other forms of coupling or no coupling at all.

QUESTION 5:

Which of the following situations represents stamp coupling between modules?

a. A module passes only the required data items to another module
b. A module passes an entire record or data structure, although only part of it is needed
c. One module directly modifies the internal logic of another module
d. A module passes control information that affects the behavior of another module
e. Two modules share access to the same global data structure
f. A module receives a complete student structure but uses only the roll number field

Correct Answer: b. A module passes an entire record or data structure, although only part of it is needed
f. A module receives a complete student structure but uses only the roll number field

Detailed Solution:

Stamp coupling occurs when a module passes an entire data structure (record/object) to another module, but only a portion of it is actually used—as in passing a full student record but using only the roll number field (f). Passing only required data items (a) is data coupling, direct logic modification (c) is content coupling, passing control flags (d) is control coupling, and sharing global data (e) is common coupling.

QUESTION 6:

A good modular software design generally aims to achieve which of the following characteristics?

a. High cohesion among the elements within a module
b. Low coupling between different modules
c. Extensive dependence among different modules
d. Modules performing closely related functions
e. Strong interaction and interdependence among all modules
f. Minimal unnecessary communication between modules

Correct Answer: a. High cohesion among the elements within a module
b. Low coupling between different modules
d. Modules performing closely related functions
f. Minimal unnecessary communication between modules

Detailed Solution:

Good modular design aims for high cohesion (elements within a module are closely related), low coupling (minimal dependence between modules), modules that perform closely related functions, and minimal unnecessary communication between modules. Extensive dependence (c) and strong interaction among all modules (e) are signs of poor design—they indicate high coupling and low independence.

QUESTION 7:

Which of the following statements are related to coupling in software design?

a. Coupling indicates the degree of interdependence between modules
b. High coupling is generally desirable in modular software design
c. The amount of data exchanged between modules affects coupling
d. Coupling measures how closely the internal elements of a module are related
e. Low coupling improves module independence and maintainability
f. Modules with excessive interaction are usually considered highly coupled

Correct Answer: a. Coupling indicates the degree of interdependence between modules
c. The amount of data exchanged between modules affects coupling
e. Low coupling improves module independence and maintainability
f. Modules with excessive interaction are usually considered highly coupled

Detailed Solution:

Coupling measures interdependence between modules; it is affected by the amount of data exchanged, and low coupling improves independence and maintainability. Excessive interaction indicates high coupling. High coupling is not desirable (b), and coupling does not measure internal relatedness—that is cohesion (d).

QUESTION 8:

A module whose activities are related because they must occur within the same period of time exhibits which of the following types of cohesion?

a. Temporal cohesion
b. Functional cohesion
c. Sequential cohesion
d. Coincidental cohesion
e. Procedural cohesion
f. Logical cohesion

Correct Answer: a. Temporal cohesion

Detailed Solution:

Temporal cohesion occurs when module elements are grouped together because they are executed at the same time (e.g., initialization, shutdown, month-end processing)—not because they are functionally related. Functional, sequential, procedural, and logical cohesion describe other grouping criteria, while coincidental cohesion is the weakest form with completely unrelated elements.

QUESTION 9:

If coupling between two modules cannot be completely avoided, which two of the following types of coupling are considered the most desirable?

a. Data coupling
b. Control coupling
c. Common coupling
d. Stamp coupling
e. Content coupling

Correct Answer: a. Data coupling
d. Stamp coupling

Detailed Solution:

Data coupling (sharing only simple, necessary parameters) is the most desirable, followed by stamp coupling (passing a structure but using only part of it)—both keep interdependence low. Control, common, and content coupling are progressively tighter and less desirable, with content coupling being the worst.

QUESTION 10:

Which three of the following types of cohesion are considered weak forms of cohesion in software design compared to other forms of cohesion?

a. Coincidental cohesion
b. Functional cohesion
c. Temporal cohesion
d. Communicational cohesion
e. Logical cohesion
f. Sequential cohesion

Correct Answer: a. Coincidental cohesion
c. Temporal cohesion
e. Logical cohesion

Detailed Solution:

Coincidental (random grouping), temporal (time-based grouping), and logical (similar but unrelated tasks) cohesion are the weakest forms—they make modules harder to understand and maintain. Functional, sequential, and communicational cohesion are stronger and more desirable.

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